

Commonly Encountered Fractures in Sports Medicine

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Almost any type of fracture can occur during athletic activities. This is because of the many different types of sports and the corresponding situations in which athletes are found. All athletes undertake various fracture risks with participation. Many of these specific fractures are covered in detail in other chapters. This chapter focuses on certain specific fractures and their management, particularly on stress fractures in various bones in the extremities and spine. Also discussed are some of the more commonly seen acute fractures in sports, such as clavicular and rib fractures. Returning an injured athlete to practice and competition, whether a recreational or an organized sport, is an issue specific to athletes. Return to participation guidelines are addressed and guidance for practicing sports medicine clinicians is offered further on.

STRESS FRACTURES

Stress fractures are commonly seen in athletes and in the military, particularly in trainees. They are seen in as many as 20% of military trainees¹ and in up to 20% of individuals presenting to a sports medicine clinic.² History of a prior bone stress injury is the strongest predictor of a new injury.^{3,4} Athletes are susceptible to developing stress fractures because they are often exposed to repetitive high-intensity, load-bearing activities in both training and competition. This overuse leads to microdamage to the involved bone that is not repaired by normal bone homeostasis. These microcracks accumulate and lead to fatigue failure and eventually a visible fracture line can be seen on imaging.⁵ Stress injuries occur on a continuum from an asymptomatic stress reaction visible only on magnetic resonance imaging (MRI) as edema, to a symptomatic fracture line visible on imaging, to a chronic nonunion. Unfortunately physical exam including the use of a tuning fork and ultrasound are of little use for diagnosing a stress fractures.⁶ Both bone scan and MRI are commonly used for imaging diagnosis, although MRI is more useful; it involves comparatively less radiation and testing time, fewer false positives, and better correlation with return to participation.^{7,8} A recently proposed classification system (Table 10.1) has been found to have high intra- and interobserver reliability and was also found to be easily remembered.⁹

Certain bone locations are more susceptible to stress injuries. Slower bone remodeling occurs in cortical bone; therefore cortical regions are more likely to sustain stress fractures. On the other hand, cancellous bone has a higher turnover and is

susceptible to altered homeostasis from poor bone quality (i.e., osteoporosis). Typically cancellous bone stress injuries occur in athletes with low bone mineral density (BMD). Furthermore, the location of the fracture within the bone can be predictive of healing potential. Injuries on the compression side of a bone are likely to do well with nonoperative management. Those that occur on the tension side of a bone are inherently unstable and may require surgery to heal.⁵ Some specific bone locations, such as the fifth metatarsal and navicular, are also predisposed to stress injury due to poor vascularity. These watershed areas are likely less able to respond to mechanical stress with an appropriate and robust healing response.¹⁰

Female Athlete Triad

Female sex is the most important predictor of stress fracture in endurance athletes. Long-distance running, in particular, poses the highest risk; it can predispose to stress fractures by itself. However, associated nutrition and menstrual irregularities are likely major contributors to injury. Lower levels of estrogen in the amenorrheic female athlete may be due to lower levels of gonadotropins. A relative lack of estrogen disrupts the normal bone turnover response to repetitive stress.⁵

The female athlete triad involves low energy availability, menstrual dysfunction, and low BMD. Early intervention is important to prevent the development of eating disorders, amenorrhea, and osteoporosis. Cardiovascular health as well as endocrine, gastrointestinal, renal, and neuropsychiatric health can also be affected.¹¹ There is evidence that athletic performance may worsen in athletes with triad symptoms.¹² Athletes can be stratified as at low, moderate, and high risk based on specific criteria involving the presence of a dietary restriction, low body mass index (BMI), delayed onset of menarche, other menstrual abnormalities, low Z score on DEXA scan, and the presence of a bone stress injury.¹¹ Female athletes who fall into the moderate- and high-risk categories are two and four times more likely to develop a bone stress injury.¹³

Biomechanical Factors

Biomechanical factors may also be associated with stress fracture risk. Interestingly, ground reactive forces do not seem to be different between uninjured and injured athletes.¹⁴ Increased lower body muscle mass may help to absorb forces during running. Athletes with smaller calf girth and less lower limb muscle mass have a higher risk of injury. Lower extremity limb alignment

Abstract

Almost any type of fracture can occur in athletes owing to the many different types of sports and the corresponding situations in which they are found. All athletes undertake various fracture risks with participation. Stress fractures are commonly seen in endurance athletes and in the military. Athletes are susceptible to developing stress fractures because they are exposed to repetitive, high-intensity, load-bearing activities. Female sex and a history of prior bone stress injury are most predictive for the development of a stress injury. Low-risk fractures, such as posterior tibial stress fractures and medial femoral neck stress fractures, can be treated nonsurgically with a period of rest and progressive rehabilitation. Higher-risk stress fractures, such as those of the anterior tibial cortex and the lateral femoral neck, are treated with surgical repair. Rib fractures can be acute traumatic or a stress injuries. Both types of rib fractures in athletes are treated nonoperatively. Rest, ice, and analgesia are important while the fracture is given sufficient time to heal. As healing progresses, activities are modified and guided by symptom improvement. Clavicular fractures are common in athletes, with midshaft fractures being most common. Nondisplaced fractures are successfully treated nonsurgically. Midshaft clavicular fractures are often displaced and their optimal treatment is debated. In athletes planning for early return to play, optimal function and early union are desirable. The findings of nonoperative treatment can be unacceptable. There is a trend toward the repair of displaced midshaft fractures, with studies suggesting higher rates of union, fewer malunions, earlier return to sport, and improved function.

Keywords

stress fracture
rib fracture
clavicle fracture
return to play
female athlete triad

TABLE 10.1 Stress Fracture Classification System

Grade	Pain	Radiographic Findings (CT, MRI, Bone Scan, or Radiography)	Description
I	No	Imaging evidence of stress fracture, no fracture line	Asymptomatic stress reaction
II	Yes	Imaging evidence of stress fracture, no fracture line	Symptomatic stress reaction
III	Yes	Nondisplaced fracture line	Nondisplaced fracture
IV	Yes	Displaced fracture (≥ 2 mm)	Displaced fracture
V	Yes	Nonunion	Nonunion

CT, Computed tomography; MRI, magnetic resonance imaging.
 From Kaeding CC, Miller T. The comprehensive description of stress fractures: a new classification system. *J Bone Joint Surg Am*. 2013;95:1214–1220.

may matter as well. A quadriceps angle greater than 15 degrees is related to a higher risk of stress fractures. Also, leg-length discrepancies are related to injury risk in runners.⁵ There may be a training adaptation period during which an athlete or military recruit is at increased risk of stress fracture. After this period, the stress injury risk may decrease.¹

Fatigue and Training Errors

There is some association between athlete fatigue and changes in loading patterns in the lower extremity. This may place a fatigued athlete at higher risk of fracture. Changes in training regimen, such as sudden increases in running mileage, are also risk factors for injury.¹⁵ Running in a fatigued state may lead to greater lower extremity mechanical forces.^{16–18}

Nutrition

Nutrition and diet influence the development of stress injuries. Higher dairy and calcium intake results in fewer stress fractures.^{3,19,20} Also, a higher intake of calcium may have a protective effect when taken in doses of 1500 to 2000 mg/d.^{5,20–23} Both vitamin D intake and vitamin D levels are lower in athletes who develop stress fractures and are inversely correlated with the risk of developing a stress fracture in both the adolescent and adult population.^{24–27}

Genetic Factors

Some recent data support a genetic predisposition to the development of stress injuries. The RANK pathway has been implicated in distance athletes.²⁸ The androgen receptor gene CAG allele distribution is different among military recruits with bone stress injuries versus those without injury.²⁹ The *CALCR* and *VDR* genes may also confer a risk for stress fracture.³⁰ Other gene sequences have been implicated as well.^{31,32} To date, however, no clear cause-and-effect relationship has been determined.

The following sections focus on the specific stress fractures most commonly seen in athletes. Risk factors for injury, recommended diagnostic workup, treatment strategies, and in some cases prevention are discussed.



Fig. 10.1 A T2-weighted magnetic resonance image showing a medial femoral neck stress fracture. (From Canale ST, Beaty JH. *Campbell's Operative Orthopaedics*. Philadelphia: Mosby; 2008.)

Femoral Neck Stress Fractures

A femoral neck stress fracture can be devastating to an athlete and the long-term health of his or her hip. Care should be taken to avoid missing a rare presentation in children.³³ Fractures cause anterior groin pain of insidious onset that is worse with weight bearing. Fractures can be bilateral even with unilateral groin pain.³⁴ A missed stress fracture can lead to displacement and the development of femoral head avascular necrosis (AVN). These fractures are described in long-distance runners and military recruits, specifically those with lower levels of fitness at the start of training.³⁵ Internal forces in the femur are concentrated along the femoral neck with running,³⁶ and there may be a genetic predisposition to the development of femoral neck stress fractures.³¹ Additionally there is an association of femoroacetabular impingement, specifically acetabular overcoverage or retroversion, with the development of femoral neck stress fractures.^{37–39}

Athletes with groin pain should undergo a standard hip radiographic series. If no fracture is visualized on radiographs, then an MRI can serve to evaluate the femoral neck in more detail (Fig. 10.1). Most fractures are first diagnosed on MRI, and an abbreviated MRI sequence is effective.⁴⁰ A bone scan positive on all three phases also indicates a stress fracture, and a single photon emission computed tomography (SPECT) scan has improved diagnostic accuracy.⁴¹ However, lack of anatomic detail limits its usefulness for treatment planning. Fractures can be classified based on whether they are nondisplaced or displaced, complete or incomplete, tension side or compression side. These characteristics have implications for recommended treatment.

Displaced fractures require urgent treatment, as they are typically seen in younger patients in their 20s and 30s. Open reduction and internal fixation is required, with a goal of restoring function and decreasing the chance of developing AVN and secondary osteoarthritis. Treatment of nondisplaced fractures depends on fracture location and the extent of involvement of the femoral neck. Fractures that are incomplete and involve only the compression side (medial aspect) of the femoral neck can

be treated nonoperatively with an extended period of non-weight bearing (Fig. 10.2). Time to return to running for compression-sided injuries increases with the severity of injury as seen on MRI. This ranged from 7.5 weeks for low-grade injuries seen only as edema without a visible fracture line to 17.5 weeks when a fracture line is visible.⁷

Fractures that involve the tension side (lateral aspect) of the femoral neck are typically treated surgically (Fig. 10.3). Percutaneous internal fixation is achieved with cannulated screws to prevent varus displacement. Of note, there is no correlation between the femoral neck shaft angle and the outcome of femoral neck stress fracture treatment.⁴² Postoperatively these patients



Fig. 10.2 An anteroposterior radiograph of the proximal femur showing a compression-side femoral neck fatigue fracture. Periosteal new bone formation is present in the inferior femoral neck, and healing endosteal callus formation is present in the inferior half of the femoral neck. (From Shin AY, Gillingham BL. Fatigue fractures of the femoral neck in athletes. *J Am Acad Orthop Surg.* 1997;5:293–302.)

are also treated with an extended period of non-weight bearing. Both nonoperatively and operatively treated patients require an extended period of time for rehabilitation and return to running as well as other high-impact athletic activities. It is important to gradually increase loading during the rehabilitation process so as to be sure that the femoral neck is healed and able to handle the required stresses.

Tibia Stress Fractures

Tibia stress fractures are the most common stress injuries in endurance athletes.⁴³ Athletes place significantly more load on the tibia while running than while walking.^{44,45} It is unclear if mechanical factors alone can account for tibia stress fractures; overall bone health along with other factors likely play an important role.⁴⁶ Running speed has been correlated with risk of tibial stress fracture.⁴⁷ Although such fractures are not as devastating as some other lower extremity stress injuries, they can lead to prolonged time off from running and impact activities.

Running athletes, including military trainees, who experience focal tibial pain with impact exercise should be evaluated for a tibial stress fracture. Patients who have such fractures are tender focally over the location of the fracture. Provocative maneuvers such as hopping on the injured leg and using the leg as a fulcrum may cause pain. A stress fracture can be distinguished from medial tibial stress syndrome (MTSS) by the focal nature of the fracture compared with the classic diffuse area of tenderness in MTSS.⁴³ Radiographs are often negative, although the presence of the “dreaded black line” is a poor prognostic indicator for fracture healing with nonoperative treatment. In most cases this represents a chronic fracture nonunion. Stress injuries not visible on radiographs can be imaged with MRI. Bone scan is used less commonly for reasons listed previously.

Tibial stress fractures are most commonly located postero-medially on the compression side of the proximal tibia. Fractures can also occur on the tension side, and these are seen along the anterior cortex of the midshaft of the tibia. Those fractures occurring on the compression side of the tibia can be treated

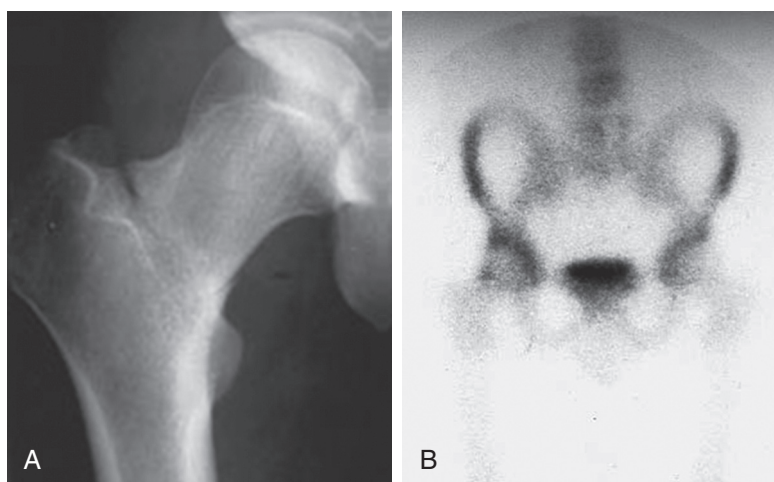


Fig. 10.3 (A) An anteroposterior radiograph with tension-side cortical disruption and propagation of the fracture across the femoral neck. (B) Magnetic resonance imaging with uptake on the superior (tension) side of the right femoral neck.



Fig. 10.4 (A and B) This tibial stress fracture is incomplete medial to lateral, but cortical disruption facilitates the diagnosis. (C) Magnetic resonance imaging demonstrates the lesion with high sensitivity. (Courtesy James W. Brodsky, MD.)

with a period of non-weight bearing and immobilization (Fig. 10.4). Fractures that occur on the tension side of the tibia and those that are visible on radiographs typically require operative intervention to heal (Fig. 10.5).⁴⁸ Traditionally these fractures were treated with an intramedullary nail. The intramedullary device promotes fracture healing by creating a load-sharing environment with the native tibia. Additionally, reaming is thought to stimulate bone marrow and provide a source of internal bone graft within the canal of the tibia to assist with healing. These fractures can also be treated with anterior tibial cortex plating with or without local bone grafting. The advantages of plating include placing the implant on the tension side of the fracture, allowing access for direct fracture débridement and bone

grafting, as well as not violating the knee joint. Such patients may experience less anterior knee pain, which is as high as 80% in some studies involving the use of an intramedullary nail to treat these injuries.⁴³ Drawbacks include plate prominence over a location of the tibia with minimal soft tissue coverage as well as the potential to devitalize the tibial cortex surrounding the fracture.^{49,50}

Proximal Fifth Metatarsal Stress Fractures

The proximal aspect of the fifth metatarsal is also a common site for a stress fracture. Athletes of any running or marching sport are susceptible. Abnormal forefoot loading may contribute to these injuries, especially with minimalist shoe wear, which

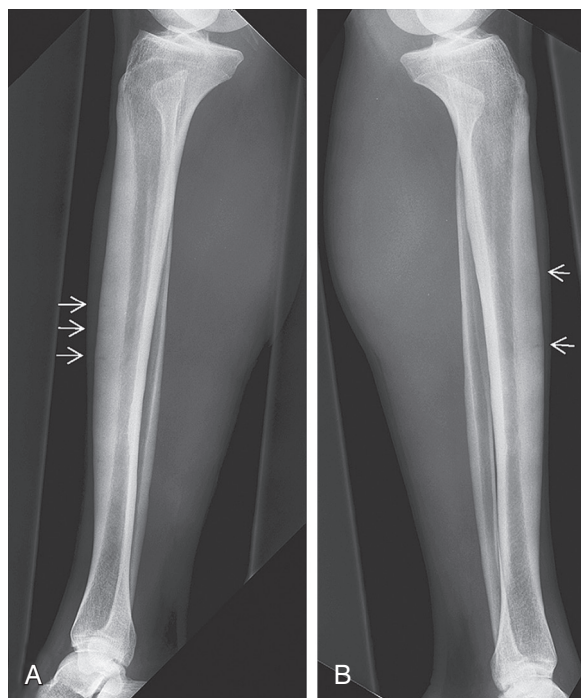


Fig. 10.5 (A and B) Radiographs of both tibiae showing multiple wedge-shaped fractures of the anterior cortex of the midtibia, the “dreaded black lines,” indicating high-risk tension-side stress fractures (arrows). (From Berger FH, de Jonge MC, Maas M. Stress fractures in the lower extremity: the importance of increasing awareness amongst radiologists. *Eur J Radiol.* 2007;62:16–26.)

increase forefoot loading.⁵¹ Athletes with a cavovarus hindfoot place additional stresses on the lateral border of the foot, predisposing them to injury. Additionally, the proximal metatarsal is a vascular supply watershed area, which perhaps leads to a poor regenerative response to repetitive injury. Also, toe grip weakness is correlated with an increased incidence of fifth metatarsal stress fractures.⁵²

These injuries should be suspected in athletes with pain along the lateral border of the foot. The pain is often insidious and chronic, although the injury may also present with pain after a twisting type of injury. Stress fractures of the proximal fifth metatarsal should be distinguished from Jones fractures, which involve the fourth and fifth metatarsal articulation, as well as smaller, proximal avulsion type injuries. This distinction can be made on plain radiographs, where stress injuries are visible more distal at the proximal junction of the metatarsal shaft and metaphysis. These injuries can be seen as a lateral cortical thickening representing attempted healing in response to repetitive stress. They may also present with a visible fracture line that most often is incomplete and involves the lateral cortex. MRI and bone scan can be used for diagnosing injuries not seen on radiographs. Ultrasound has been proposed as a cheaper imaging option, although its sensitivity is not as high as that of MRI.⁵³

When a fracture is visible on radiographs, most patients are treated with surgical stabilization (Fig. 10.6). Distinguishing a true Jones fracture from a diaphyseal fracture may not be important.⁵⁴ Nonoperative treatment has higher nonunion rates and longer time to sports and physical labor.⁵⁵ Typically surgical repair



Fig. 10.6 An acute Jones fracture at the diaphyseal-metaphyseal junction of the fifth metatarsal. (Courtesy James W. Brodsky, MD.)

involves placing an intramedullary screw with sufficient screw threads past the fracture (Fig. 10.7). Intramedullary cortical thickening may make canal drilling difficult. Additionally, it is important not to penetrate the cortex distally with the drill or screw, thus creating a stress riser susceptible to fracture. There is debate over screw diameter, since larger screws are less susceptible to fatigue fracture.⁵⁶ For most patients a screw 5 mm to 50 mm in diameter, based on computed tomographic (CT) studies, is appropriate.⁵⁷ Some surgeons prefer a solid screw as opposed to a cannulated one to also help resist fatigue failure. Most commonly, exposing the fracture site to débride the fracture and place bone graft is not necessary for a first-time attempt at fixation, although it may be considered on a case-by-case basis and might be more likely in revision cases. Patients with incomplete fractures on radiographs and a plantar fracture gap less than 1 mm healed faster than those with a gap greater than 1 mm.⁵⁸ Patients with a cavovarus hindfoot might require a valgus-producing calcaneal osteotomy to unload the lateral aspect of the foot and create a more favorable mechanical environment.⁵⁹ This is especially important in athletes who sustain recurrent injuries.

Postoperatively an assessment should be made of the factors, in addition to patient-specific health characteristics, that may have contributed to the injury. These may include running style, training volume, running shoe quality and wear, as well as foot shape. Additionally, athletes with low vitamin D levels (<30 ng/mL) have a higher incidence of fifth metatarsal stress fracture and should therefore be treated with oral supplementation.^{26,60} Adjusting each of these factors is important in order to minimize the chance of recurrence. Custom orthotics should be considered to ensure that stresses to the athlete's feet are properly distributed. Recurrence of injury after operative fixation is more likely in athletes who return to sports earlier, prior to complete healing.^{61,62} This is also true for those with a higher BMI and a wider fourth-to-fifth intermetatarsal angle and curved fifth metatarsal head.⁶³

Other Metatarsal Stress Fractures

Stress fractures involving the first to fourth metatarsals are also common, especially in running athletes and dancers.⁶⁴ The second metatarsal is subjected to the highest bending stresses and is therefore the most commonly involved of the metatarsals. This is followed by the third metatarsal.⁶⁵

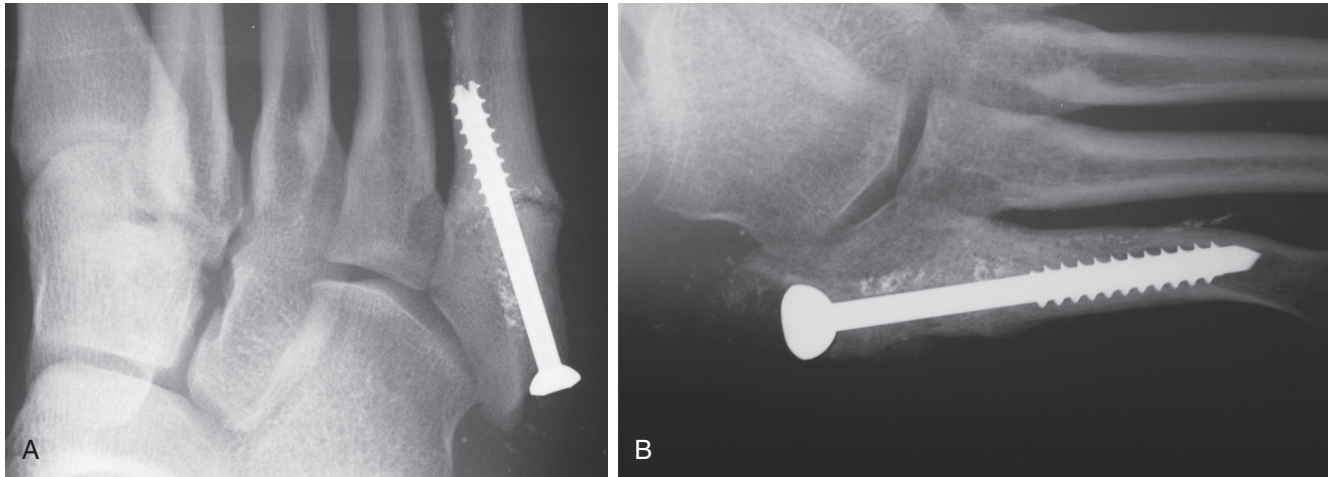


Fig. 10.7 Images of an 18-year-old collegiate soccer player with recurrent pain and nonunion of a Jones fracture after previous surgery. (A) Note the hypertrophic beaking and sclerosis at the fracture site. (B) Healing was achieved 10 weeks after revision intramedullary fixation and bone grafting. (Courtesy James W. Brodsky, MD.)

Fractures can occur anywhere along the metatarsal and maximal stress occurs 3 to 4 cm distal from the base of the bone.⁶⁵ Fractures that are more distal heal more predictably. In most cases, with a period of immobilization and limited weight bearing, treatment is successful. Once the foot pain has subsided, it is important to progress activities in a gradual process to allow for mechanical adaptation of the foot to increased loads. It is also important to address any factors involving the patient's systemic homeostasis as well as foot biomechanics and loading patterns. Achilles stretching, cross training, limiting mileage, regular shoe replacement, and custom orthotics are all treatment strategies that should be considered.

Forth metatarsal fractures can have delayed or incomplete healing with nonoperative treatment, especially in patients with metatarsus adductus.⁶⁶ Open reduction and plate fixation is an option in these injuries to hasten return to athletics.⁶⁷

Navicular Stress Fractures

Navicular stress fractures, while uncommon, are challenging to diagnose and treat. Delay in diagnosis is the norm.⁶⁸ They are seen in sprinting and jumping sports such as basketball and, unlike most stress fractures, are more common in male athletes.^{4,65} The central third of the bone is an area of high shear stress. The navicular is the keystone to the medial longitudinal arch and the link between the midfoot and hindfoot. It serves as an important structural support for both the longitudinal and transverse arches.⁶⁹ With weight bearing, forces are generated medially across the navicular from the first metatarsal through the medial cuneiform and the talus. Forces through the second metatarsal and middle cuneiform are not resisted in the same way, thereby generating shear forces. Additionally, the central third of the navicular is thought to be a relatively avascular zone, which limits its ability to resist changes in forces and subsequently its healing potential.^{10,70} Recent data note that an avascular zone in the central navicular is present only 12% of the time. The majority of the time the navicular is well perfused from the dorsalis pedis and branches of the posterior tibial artery.⁷¹

Patients present with insidious dorsal foot pain, and diagnosis of a navicular stress injury can often be delayed. Tenderness is described over the navicular. Pain can be exacerbated by axially loading a plantarflexed foot such as standing on tiptoe or hopping on a single leg.⁶⁸ Radiographs may become positive only on a delayed basis owing to lack of involvement of the plantar surface.⁶⁵ Therefore MRI is more commonly used for fracture detection.

Once the fracture has been diagnosed, treatment requires an extended period of immobilization. When fractures are diagnosed in early stages, they can be treated successfully with casting for a minimum of 6 weeks. Non-weight bearing during this period is also important; allowing weight bearing too soon can lead to nonunion.^{72,73} On average, these fractures result in about 4 months loss of athletic participation.⁷² Pain with therapeutic ultrasound can be used to follow navicular healing and correlates with MRI findings.⁷⁴

Athletes with fracture lines visible on radiographs or those hoping to return to participation sooner may be candidates for operative fixation. Operative treatment allows return to athletics at a mean of 16 weeks, compared with 22 weeks for conservative treatment.⁴⁸ In most cases two partially threaded screws placed perpendicular to the fracture line from lateral to medial suffice to generate compression and healing (Fig. 10.8). Postoperative limited weight bearing for 1 to 2 months is important. Repeat imaging to confirm fracture healing can be considered prior to release to return to sports participation. Outcomes of navicular stress fractures are not always optimal regardless of treatment. There is a high rate of persistent pain and disability and those patients are more likely to develop a nonunion.⁷⁵ With operative treatment there is a high rate of secondary osteoarthritis and need for reoperation.⁷⁶

Pars Stress Fractures

Stress fractures of the lumbar spine are commonly seen in athletes who perform repetitive lumbar extension activities. Sports-related back pain in patients under the age of 18 years is due to a stress injury up to 40% of the time.^{77,78} Although less

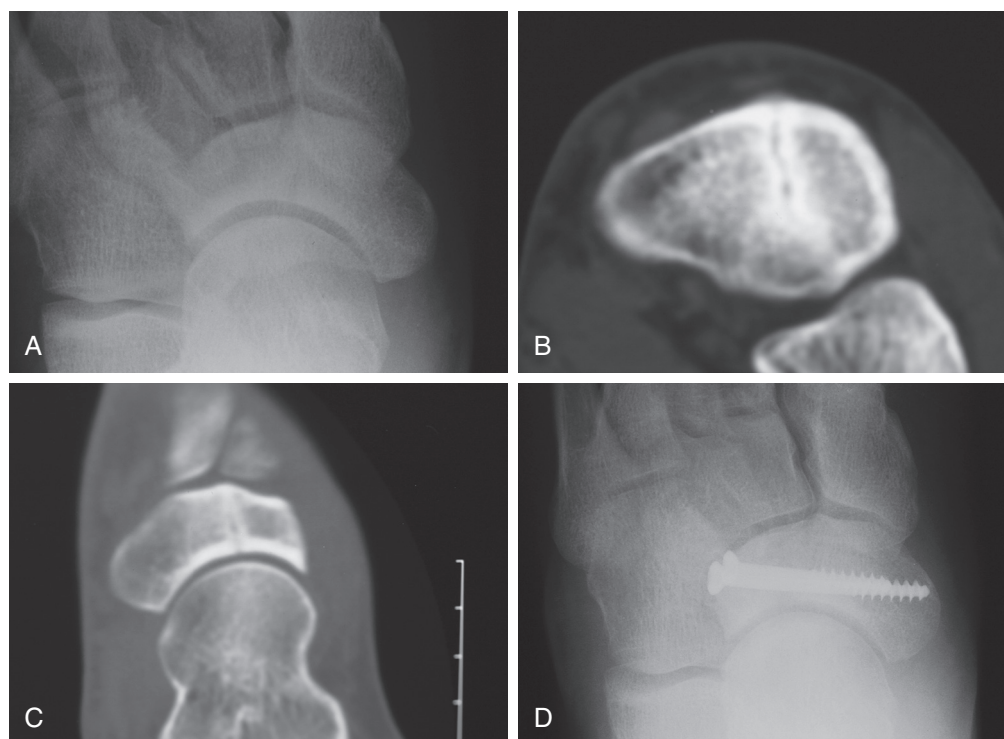


Fig. 10.8 A navicular stress fracture. (A) Subtle linear lucency in the navicular is seen on the radiograph. (B and C) Computed tomography confirmed the diagnosis. (D) The patient was treated with internal fixation and bone grafting. (Courtesy James W. Brodsky, MD.)

common, new-onset spondylolysis is also seen in adults.⁷⁹ Susceptible athletes include gymnasts and football linemen. Repetitive axial loading of the lumbar spine with spinal extension leads to stress injuries of the pars interarticularis, or spondylolysis. These injuries are usually bilateral, and most occur at L5, with L4 being the next most common location. Most patients with spondylolysis are asymptomatic, although the diagnosis of stress injury should be kept in mind with athletes presenting with chronic lumbar pain.⁸⁰

Patients presenting with spondylolysis may have a stiff-legged gait with shortened stride length due to hamstring spasm, along with limited forward flexion. A palpable step-off may be present in patients with advanced spondylolisthesis. Lumbar nerve root radiculopathy may be present as well, from irritation due to the nearby inflamed pars defect. Plain radiography is part of the initial workup, and oblique views are more sensitive for diagnosis. Flexion-extension radiographs can document and grade instability in patients with spondylolisthesis. Advanced imaging with CT is used to best evaluate the bony architecture and MRI to evaluate the nerve roots and associated disk disease (Fig. 10.9). MRI has a high sensitivity as well for diagnosing spondylolysis and is likely comparable to CT.^{81,82} SPECT bone scan is the most sensitive test for diagnosis.⁸⁰

Patients with pars defects only and those with low-grade spondylolisthesis are typically treated with activity modification and extension bracing. Rehabilitation involving lumbopelvic flexibility and strengthening is important in the later stages of recovery and to foster return to participation. Additionally, stress injuries to the pars interarticularis are seen in patients with low vitamin D levels; therefore testing should be considered.²⁵ Most

patients will improve with nonoperative treatment; however, the union rate is low and only around 25% heal successfully.⁸³ Additionally, patients are susceptible to recurrence of symptoms as long as they remain involved in high-level athletics.⁸⁴ Patients unresponsive to extensive conservative treatment, those with neurologic deficits, and patients with unstable and high-grade spondylolisthesis may consider pars repair if the associated disks are nondegenerative. In patients with degenerative disk disease, a lumbar fusion of the involved levels is indicated. However, the outcomes data are limited for adult athletes who have failed nonoperative treatment. Surgical pars interarticularis repair is likely to bring improvements in pain, function, and eventual return to athletics.⁸⁵ Studies on return to play (RTP) after pars repair note that this often takes 1 year from the time of surgery, especially for contact sports, and is successful around 80% of the time.⁸⁶

Rib Fractures

Rib fractures in athletes can be troublesome. They can either be traumatic or nontraumatic secondary to stress injury. Although traumatic injuries may be easier to diagnose, stress fractures of the ribs can be missed and present variably in athletes. It is important for the clinician to be familiar with the risks and causes of stress fractures and to have a high level of suspicion in the athlete presenting with atraumatic pain.

Traumatic injuries are typically easy to diagnose, since they are usually associated with a direct blow to the rib cage. There will be very localized pain at the fracture site, which is exacerbated with direct pressure or deep breaths. On examination, a deformity may be seen, with a hematoma or overlying bruising.

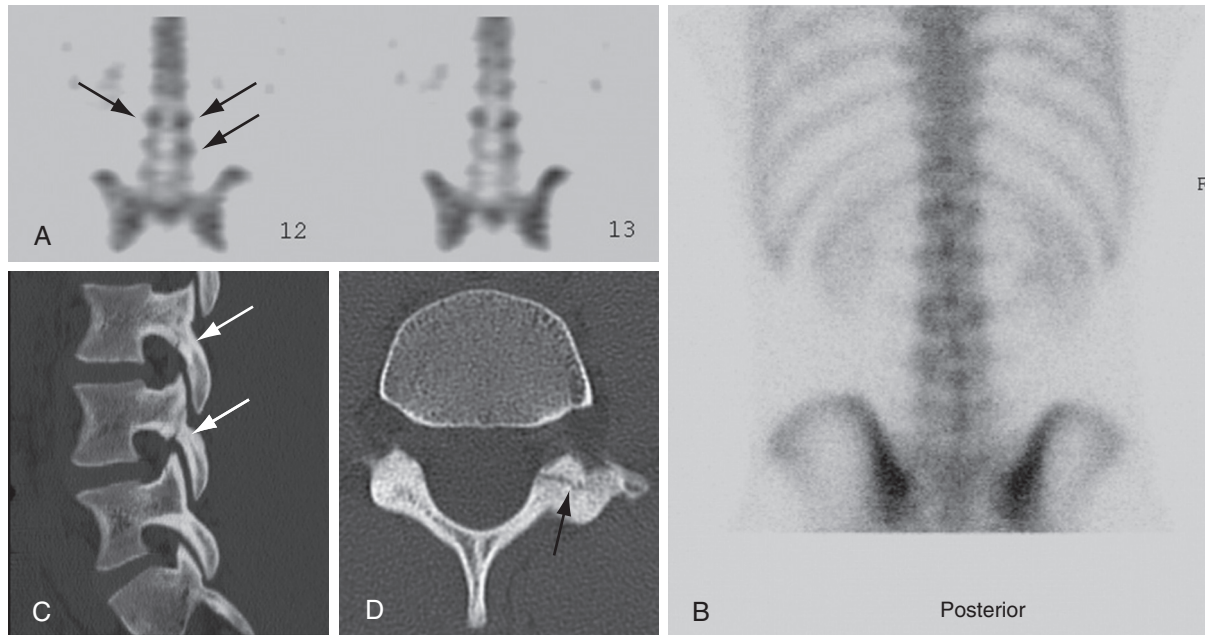


Fig. 10.9 A pars stress fracture. (A) Four-hour delayed bone single-photon emission computed tomography (CT) images of the lumbar spine demonstrating increased uptake in the bilateral L3 posterior elements and the right L4 posterior element corresponding to active pars interarticularis stress fractures (*black arrows*). (B) The abnormalities are barely visible on the standard bone scan image of the lumbar spine. (C) A sagittal reformation CT image demonstrates linear lucencies in the right L3 and L4 pars interarticularis corresponding to areas of abnormal uptake (*white arrows*). (D) An axial CT image demonstrates a left-sided pars defect at L3 (*black arrow*).

The physician is typically able to localize the area of pain with palpation directly over the fracture. Deep breaths and any motions requiring contraction of the thoracic musculature will cause pain. Anteriorly, separation of the costochondral junction is possible with intense thoracic contraction and twisting motions. Such injuries are common in wrestlers, for example. With worse separations, the displacement at the costochondral junction can be seen and felt.

Stress fractures of the ribs are much less common, present variably, and can often be missed in the differential diagnosis. Symptom onset is usually insidious, and athletes will typically have pain only during the inciting activity. In the early stages they may complain of shoulder or abdominal pain, depending on the location of the fracture. The pain will worsen as the athlete continues with the causative activity without rest or modification. Rib stress fractures are primarily caused by repetitive tensile muscular forces, which are typically seen in sports requiring throwing and swinging of the upper extremities.⁸⁷⁻⁹⁴ Common sports in which stress rib fractures occur include rowing, gymnastics, basketball, and baseball. The most common sites for stress fractures include the first rib anterolaterally, the fourth through ninth ribs posterolaterally, and the upper ribs posteromedially.⁹² Generally first-rib fractures are seen in sports requiring repetitive overhead activities, such as baseball, basketball, and tennis.⁸⁸⁻⁹⁰ Athletes may present with a vague history of pain in the posterior shoulder, midclavicular region, or upper back.⁸⁹ Lower-rib stress fractures are seen in sports requiring repetitive torsion of the body, such as rowing or golfing.^{92,93}

Evaluation of the athlete with a rib fracture should include examination of the chest, heart, lungs, and abdomen. Particularly in the case of traumatic fractures, other associated organ injuries—involving the lungs, liver, spleen, or kidneys—should be ruled out. In atraumatic fractures, other potential nonmusculoskeletal causes of pain should be considered, as these patients may present variably.

Imaging studies typically begin with plain radiographs of the chest and ribs. Traumatic fractures should be visible, but imaging of the ribs can be challenging and requires careful examination of each rib. It is helpful to review the findings with a radiologist to ensure that all the fractures are noted. Stress fractures, on the other hand, are not always seen initially. Up to two-thirds of initial x-rays are negative and only become positive once healing has begun.⁹⁵

Bone scans will show increased uptake in the area of the stress fracture with 100% sensitivity and allow for earlier diagnosis versus radiographs.⁹⁶ However, MRI is the modality of choice to diagnose stress fractures. It has the highest combined sensitivity and specificity and adds the benefit of evaluating for soft tissue abnormalities.^{95,96}

Both traumatic and nontraumatic rib fractures in athletes can be treated nonoperatively. Rest, ice, analgesia, and cessation of the inciting activity are important while the fracture is given sufficient time to heal. As healing progresses, activities are modified and increasingly guided by symptom improvement. For severe pain, “rib strapping” or intercostal nerve blocks can be considered. Rehabilitation should include evaluation of mechanics

and training regimens to prevent future fractures. Nonunions of rib fractures is rare.^{87,91}

Clavicle Fractures

Clavicular fractures account for up to 10% of all sport-related fractures,^{97,98} and 30% of all clavicular fractures occur during sports.⁹⁸ These are common fractures in the athlete, with midshaft fractures being the most common (Fig. 10.10). Midshaft fractures are displaced nearly 50% of the time and comminuted in 19%.^{99,100} These injuries usually occur with direct blows to the shoulder, either from collisions with other players or falls onto the shoulder. A deformity may be seen, and the clinician should note any open wounds or skin tenting. A thorough neurovascular examination should be performed, and other concomitant injuries should be ruled out.

The treatment of clavicular fractures has been the topic of much discussion. Nondisplaced fractures can be treated successfully without surgery. However, midshaft clavicular fractures are often displaced,^{99,100} and in such cases their optimal treatment has been much debated. Traditionally all clavicular fractures were treated nonoperatively.^{101,102} However, recent reports suggest that conservative treatments may give suboptimal results, including unacceptable nonunion rates, increased return times to sports, and poor residual shoulder function.¹⁰³⁻¹⁰⁶ Particularly in the athlete, where early RTP, optimal function, and early union are desirable, the findings from nonoperative treatment are unacceptable to most. Hence there has been a trend towards surgical fixation of displaced midshaft clavicular fractures, especially with studies suggesting lower rates of nonunion and malunion, earlier return to sports, and improved functional outcomes.¹⁰⁷⁻¹⁰⁹

Surgical treatment of displaced clavicular fractures in the pediatric population remains controversial. As children increasingly

participate in sports, the optimal treatment has also been a topic of discussion for the same reasons as in the adult athlete. Because of the great remodeling potential, clavicular fractures in children have traditionally been treated nonoperatively. There are reports, however, of improved outcomes with plate fixation, particularly in adolescent patients.¹¹⁰ In younger patients, however, surgical fixation becomes more controversial. The benefits of surgical fixation should always be weighed against the risks of surgery, including infection and neurovascular injury.

Clavicular stress fractures are rare but may occur in athletes participating in rowing, diving, baseball, weight lifting, and gymnastics.^{88,91,111} These fractures are thought to occur due to a muscular imbalance of repetitive bending, shearing, and rotational forces across the clavicle.⁹¹ As most stress fractures are undisplaced, treatment is typically nonsurgical.

Most patients will successfully return to sports after a clavicular fracture, with at least 80% of patients returning to their preinjury level of play.^{109,112} The decision to return is multifactorial and begins with clinical and radiographic evidence of fracture healing. Patients should demonstrate full pain-free range of motion (ROM) with near full strength. An RTP training regimen should be started and gradually advanced before exposure to a real game situation. Sport-specific drills and functional exercises should be incorporated to make sure that the patient does not have pain and to give him or her an opportunity to regain confidence after a prolonged absence from playing. The timing of RTP is controversial and quite variably reported.^{109,113} Generally, with nonoperative treatment, return to sports may take 6 to 12 weeks from the time of injury.^{103,114,115} Surgical treatment may allow for earlier return to sport, with some reporting returns earlier than 6 weeks after surgery.^{112,113} This is a minority, however, and it appears that average return times are 68 to 83 days after surgery.^{112,113} Ultimately the decision to RTP will depend on the type of treatment, evidence of healing, and certainly by the level and type of sport (i.e., contact vs. noncontact) to which the athlete is returning. Return to sport is discussed in further detail in the next section.

RETURN TO SPORT

The decision of returning an athlete to sports participation can be complex and difficult, yet it remains a critical component of the clinician's treatment of the injured athlete. In today's competitive climate, the physician is burdened with questions of when an athlete will eventually RTP. These questions are unavoidable and come from the athlete, his or her parents, agents, coaches, and the media, to list a few. Despite being a hallmark of the clinician's treatment, the decision to RTP remains controversial and a topic of much debate. The complex nature of the decision-making process perhaps accounts for the scarcity of higher-level evidence in the literature providing a systematic and objective approach to returning an athlete to sports. The ultimate goal is to expedite the athlete's recovery and return to sports while making sure that the initial injury is healed and the risk of reinjury is minimal.

To help with the decision-making process, Creighton et al. presented a three-step model that outlines the factors to be

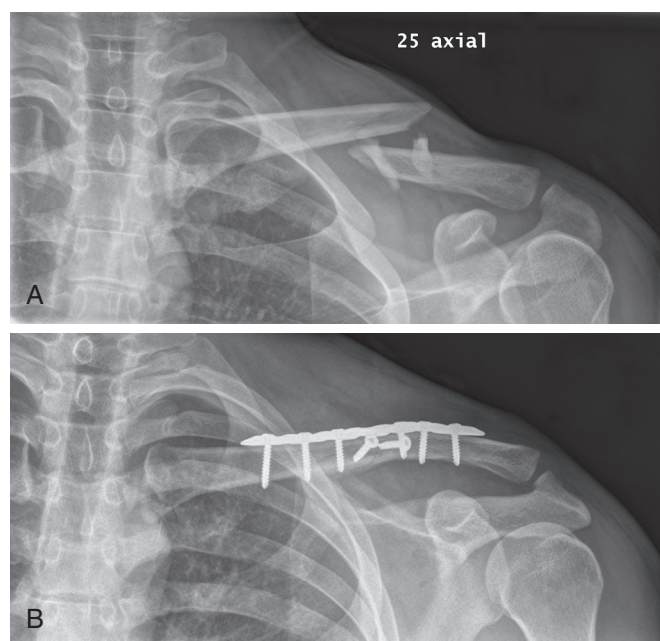


Fig. 10.10 Left comminuted midshaft clavicle. (A) Preoperative. (B) Postoperative.

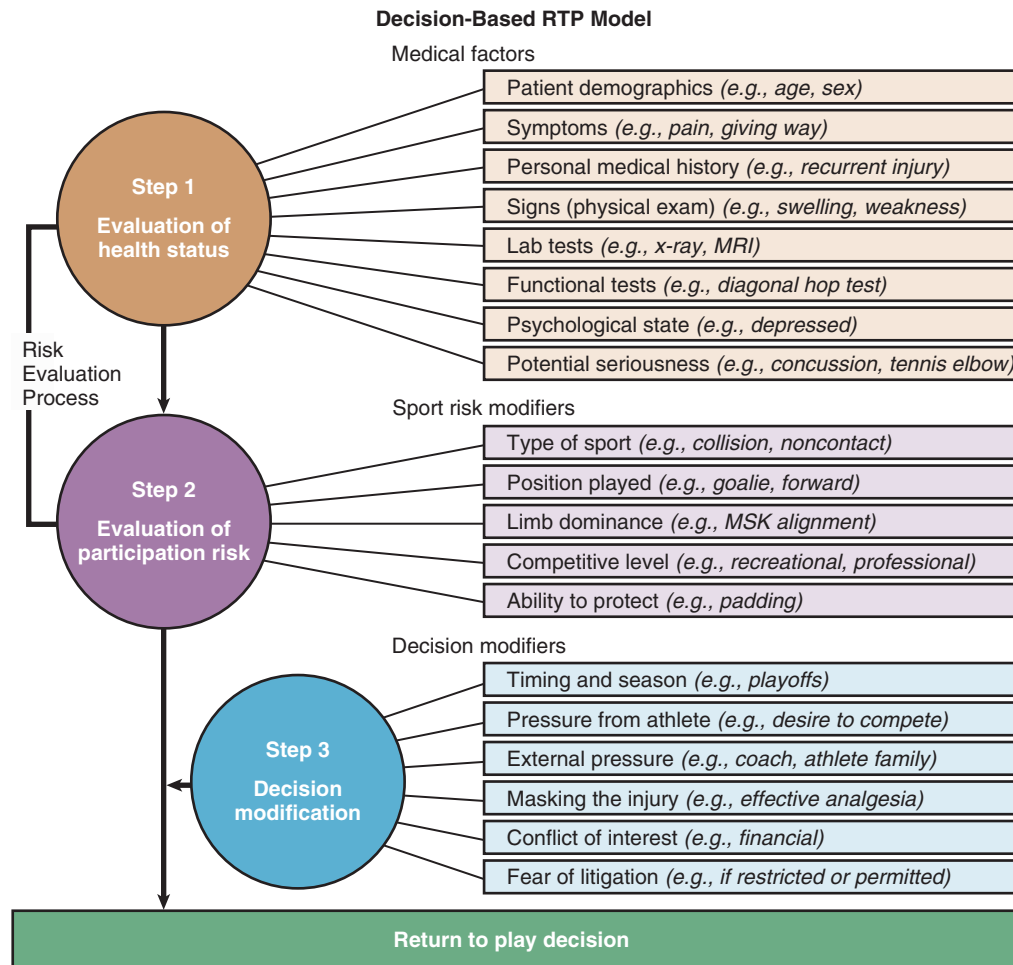


Fig. 10.11 A decision-based return to play (RTP) model. MRI, Magnetic resonance imaging; MSK, musculo-skeletal. (From Creighton DW, Shrier I, Shultz R, et al. Return-to-play in sport: a decision-based model. *Clin J Sport Med.* 2010;20[5]:379–385.)

considered before returning the athlete to play.¹¹⁶ This model incorporates important factors that go beyond the medical status of the player's injury. It includes the evaluation of the health status, evaluation of participation risk, and includes modifiers that could affect the decision (Fig. 10.11).

The first step is an evaluation of the athlete's health status. This includes typical objective and subjective medical factors that the clinician usually considers before returning an athlete to play. Much of this evaluation is done in the clinic setting, and the physician must decide whether the initial injury has healed enough to tolerate the forces of the sporting activity. The patient should have resolution of symptoms such as pain and instability. Objective assessment includes examination for ROM, laxity, strength, and the presence of swelling and inflammation. These are typically compared with the contralateral extremity, and many have established specific measurable criteria to determine readiness.^{117–119} A more dynamic and functional assessment is usually part of the decision-making process (e.g., the diagonal hop test) and includes tests to ensure adequate balance and proprioception. In many instances, these assessments are performed as part of the athlete's functional rehabilitation, with the report made available to the physician. In today's competitive

sporting climate and increasing sports specialization involving young athletes, it is common to introduce sport-specific functional rehabilitation as part of the return-to-sport process. For example, ballet dancers place different demands on their knees compared with golfers or soccer players. Therefore functional assessments would also have to be sport-specific.

Traditionally, the decision to allow athletes to RTP exclusively involved step 1 of this model. Many RTP guidelines in the literature are in fact restricted to step 1 as well.¹²⁰ According to Creighton et al., the second step is to evaluate participation risk and consider sport risk modifiers.¹¹⁶ These are factors such as the type of sport, the position played, and the competitive level of play. For example, swimming may put different demands on the knee compared with a full-contact sport such as football. A long-distance track-and-field runner with a shoulder injury may return sooner than a tennis player with the same injury. The position played also makes a difference, as a soccer goalkeeper may have different demands compared with a striker. The level of competition should also be a consideration. A weekend recreational athlete will have different demands than the professional elite athlete. Such factors should certainly come into consideration in determining an athlete's return to their sport.

The final step involves the consideration of decision modifiers and should not be taken lightly. These modifiers may be reason enough for the athlete to decide to forego surgery and continue playing through an injury or even to RTP early despite the risks of reinjury. These decision modifiers can change the decision to return if participation risk is considered alone. For example, athletes may injure themselves midseason of their high school senior year and may choose to play through the injury and delay surgery so that they may finish playing in their final year. Removing athletes from play at a critical time of the season may cause them to miss an opportunity for a college scholarship, again swaying the decision to return early. The physician may also encounter external pressure from the coach, media, and family to proceed with a certain type of treatment plan, while the athlete may also encounter the same pressures. Fear of litigation may also sway one's decision. If an athlete is not allowed to play, he may blame the physician for losing the opportunity for a bonus, scholarship, or position on a team. On the other hand, if the athlete returns to play and is reinjured, the physician may be blamed for clearing him too early. The consideration of step 3 of this model involves clear and open communication between the athlete and physician regarding appropriate expectations. The discussion may involve family, agents, coaches, and a sporting organization, but the physician must use caution when disclosing personal, medical, or other sensitive information to

different parties. These “decision modifiers” can easily be ignored by the treating physician, but their consideration is paramount in the optimal treatment of today's athletes.

Despite the increasing discussion of RTP in the literature, there are few guidelines providing a standardized approach to these difficult decisions. Creighton et al. provided one such model¹¹⁶; however, even this was later modified to address limitations in its ability to be applied in all situations.¹²¹ Some of the challenges were the model's ability to be applied to serious injuries such as concussions, or when there were multiple injuries and risks. The author proposes that these difficulties arose because the original model was a sociologic construct, whereas the decision to RTP involves biologic considerations. Therefore the Strategic Assessment of Risk and Risk Tolerance, or StARRT, framework was introduced (Fig. 10.12).¹²¹ In short, the original model was modified to produce a better fit with the biological considerations when assessing RTP. The “medical factors” were modified to “tissue health,” and the “sport risk modifiers” were changed to represent the stresses applied to the tissues. These again encompass steps 1 and 2, respectively, and represent the risk assessment process. Finally, the “decision modifiers” were changed to “risk tolerance modifiers.” Ultimately, the decision to RTP is based on whether the risk tolerance exceeds the risk assessment. In other words, if the risk assessment is greater than the risk tolerance, the athlete should not RTP.

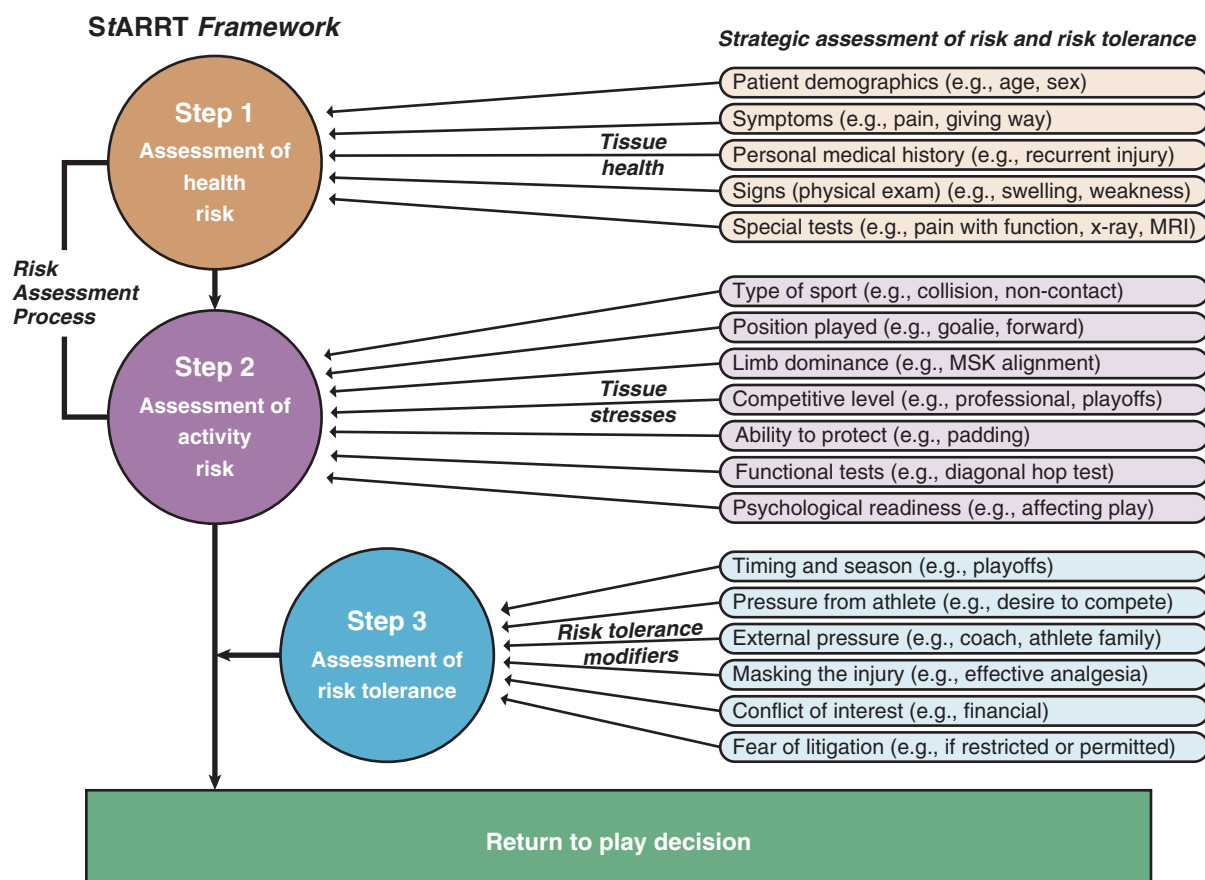


Fig. 10.12 The Strategic Assessment of Risk and Risk Tolerance (StARRT) framework for return to play. MRI, Magnetic resonance imaging; MSK, musculoskeletal. (From Shrier I. Strategic Assessment of Risk and Risk Tolerance [StARRT] framework for return-to-play decision-making. *Br J Sports Med.* 2015;49:1311–1315.)

Numerous factors are involved in the decision to return an athlete to play. Even for experienced team physicians, the decision can be complex and difficult. With the multitude of factors to be considered, however, there is still great variability among physicians with regard to the decision-making process.¹²² Ultimately the decision to RTP should be made jointly between the physician and athlete, with appropriate communication also involving other parties such as therapist, family, coaches, and the sports organization. Too much communication and involvement of outside parties may complicate the decision-making process; on the other hand, it may also help the physician to consider the multiple factors involved in deciding to return an athlete to his or her sport.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Burgi AA, Gorham ED, Garland CF, et al. High serum 25-hydroxyvitamin D is associated with a low incidence of stress fractures. *J Bone Miner Res*. 2011;26(10):2371–2377.

Level of Evidence:

II

Summary:

In this study there was a monotonic inverse dose-response gradient between serum 25(OH)D and risk of stress fracture, indicating that there was double the risk of stress fractures of the tibia and fibula in women with serum 25(OH)D concentrations of less than 20 ng/mL versus those with concentrations of 40 ng/mL or greater. A target for prevention of stress fractures would be a serum 25(OH)D concentration of 40 ng/mL or greater, achievable with 4000 IU/d of vitamin D3 supplementation.

Citation:

Tenforde AS, Carlson JL, Chang A, et al. Association of the female athlete triad risk assessment stratification to the development of bone stress injuries in collegiate athletes. *Am J Sports Med*. 2017;45(2):302–310.

Level of Evidence:

III

Summary:

Using published guidelines, 29% of female collegiate athletes in this study were classified into moderate- or high-risk categories using the Female Athlete Triad Cumulative Risk Assessment Score. Moderate- and high-risk athletes were more likely (two and four times, respectively, compared with low risk) to subsequently sustain a bone stress injury (BSI); most BSIs were sustained by cross-country runners.

Citation:

Kaeding CC, Miller T. The comprehensive description of stress fractures: a new classification system. *J Bone Joint Surg Am*. 2013;95(13):1214–1220.

Level of Evidence:

III

Summary:

This paper proposes a five-tier stress fracture classification system based on imaging and symptoms; the system is clinically relevant, easily applied, generalizable, and has excellent interobserver and intraobserver reliability.

Citation:

Miller TL, Harris JD, Kaeding CC. Stress fractures of the ribs and upper extremities: causation, evaluation, and management. *Sports Med*. 2013;43:665–674.

Level of Evidence:

III

Summary:

This paper presents a review of the literature discussing upper extremity stress fractures, including their risk factors, clinical presentation, and treatment.

Citation:

McKee RC, Whelan DB, Schemitsch EH, et al. Operative versus nonoperative care of displaced midshaft clavicular fractures: a meta-analysis of randomized clinical trials. *J Bone Joint Surg Am*. 2012;94:675–684.

Level of Evidence:

I

Summary:

This paper presents a meta-analysis of clinical trials comparing operative and nonoperative treatment of displaced midshaft clavicular fractures. The report concludes that operative treatment results in lower rates of nonunion and symptomatic malunion as well as earlier functional return.

Citation:

Shrier I. Strategic assessment of risk and risk tolerance (StARRT) framework for return-to-play decision making. *Br J Sports Med*. 2015;49:1311–1315.

Level of Evidence:

IV

Summary:

This article presents a revised framework to guide RTP decision-making. It addresses limitations noted in the original three-step model presented by Creighton et al. in 2010.

REFERENCES

- Finestone A, Milgrom C, Wolf O, et al. Epidemiology of metatarsal stress fractures versus tibial and femoral stress fractures during elite training. *Foot Ankle Int.* 2011;32:16–20.
- Field AE, Gordon CM, Pierce LM, et al. Prospective study of physical activity and risk of developing a stress fracture among preadolescent and adolescent girls. *Arch Pediatr Adolesc Med.* 2011;165:723–728.
- Tenforde AS, Sayres LC, McCurdy ML, et al. Identifying sex-specific risk factors for stress fractures in adolescent runners. *Med Sci Sports Exerc.* 2013;45:1843–1851.
- Wright AA, Taylor JB, Ford KR, et al. Risk factors associated with lower extremity stress fractures in runners: a systematic review with meta-analysis. *Br J Sports Med.* 2015;49:1517–1523.
- McCormick F, Nwachukwu BU, Provencher MT. Stress fractures in runners. *Clin Sports Med.* 2012;31:291–306.
- Schneiders AG, Sullivan SJ, Hendrick PA, et al. The ability of clinical tests to diagnose stress fractures: a systematic review and meta-analysis. *J Orthop Sports Phys Ther.* 2012;42:760–771.
- Ramey LN, McInnis KC, Palmer WE. Femoral neck stress fracture: Can MRI grade help predict return-to-running time? *Am J Sports Med.* 2016;44:2122–2129.
- Nye NS, Covey CJ, Sheldon L, et al. Improving diagnostic accuracy and efficiency of suspected bone stress injuries. *Sports Health.* 2016;8:278–283.
- Kaeding CC, Miller T. The comprehensive description of stress fractures: a new classification system. *J Bone Joint Surg Am.* 2013;95:1214–1220.
- Shindle MK, Endo Y, Warren RF, et al. Stress fractures about the tibia, foot, and ankle. *J Am Acad Orthop Surg.* 2012;20:167–176.
- De Souza MJ, Nattiv A, Joy E, et al. 2014 Female Athlete Triad Coalition consensus statement on treatment and return to play of the female athlete triad: 1st International Conference held in San Francisco, CA, May 2012, and 2nd International Conference held in Indianapolis, IN, May 2013. *Clin J Sport Med.* 2014;24:96–119.
- Vanheest JL, Rodgers CD, Mahoney CE, et al. Ovarian suppression impairs sport performance in junior elite female swimmers. *Med Sci Sports Exerc.* 2014;46:156–166.
- Tenforde AS, Carlson JL, Chang A, et al. Association of the female athlete triad risk assessment stratification to the development of bone stress injuries in collegiate athletes. *Am J Sports Med.* 2017;45:302–310.
- Zadpoor AA, Nikooyan AA. The relationship between lower-extremity stress fractures and the ground reaction force: a systematic review. *Clin Biomech (Bristol, Avon).* 2011;26:23–28.
- Hreljac A. Etiology, prevention, and early intervention of overuse injuries in runners: a biomechanical perspective. *Phys Med Rehabil Clin N Am.* 2005;16:651–667, vi.
- Derrick TR, Dereu D, McLean SP. Impacts and kinematic adjustments during an exhaustive run. *Med Sci Sports Exerc.* 2002;34:998–1002.
- Verbitsky O, Mizrahi J, Voloshin A, et al. Shock transmission and fatigue in human running. *J Appl Biomech.* 1998;14:300–311.
- Voloshin AS, Mizrahi J, Verbitsky O, et al. Dynamic loading on the human musculoskeletal system – effect of fatigue. *Clin Biomech (Bristol, Avon).* 1998;13:515–520.
- Kelsey JL, Bachrach LK, Procter-Gray E, et al. Risk factors for stress fracture among young female cross-country runners. *Med Sci Sports Exerc.* 2007;39:1457–1463.
- Nieves JW, Melsop K, Curtis M, et al. Nutritional factors that influence change in bone density and stress fracture risk among young female cross-country runners. *PM R.* 2010;2:740–750, quiz 794.
- Lappe J, Cullen D, Haynatzki G, et al. Calcium and vitamin d supplementation decreases incidence of stress fractures in female navy recruits. *J Bone Miner Res.* 2008;23:741–749.
- Tenforde AS, Sayres LC, Sainani KL, et al. Evaluating the relationship of calcium and vitamin D in the prevention of stress fracture injuries in the young athlete: a review of the literature. *PM R.* 2010;2:945–949.
- Wentz L, Liu PY, Ilich JZ, et al. Dietary and training predictors of stress fractures in female runners. *Int J Sport Nutr Exerc Metab.* 2012;22:374–382.
- Dao D, Sodhi S, Tabasinejad R, et al. Serum 25-hydroxyvitamin d levels and stress fractures in military personnel: a systematic review and meta-analysis. *Am J Sports Med.* 2015;43:2064–2072.
- McClellan JW 3rd, Vernon BA, White MA, et al. Should 25-hydroxyvitamin D and bone density using DXA be tested in adolescents with lumbar stress fractures of the pars interarticularis? *J Spinal Disord Tech.* 2012;25:426–428.
- Shimasaki Y, Nagao M, Miyamori T, et al. Evaluating the risk of a fifth metatarsal stress fracture by measuring the serum 25-hydroxyvitamin D levels. *Foot Ankle Int.* 2016;37:307–311.
- Sonneville KR, Gordon CM, Kocher MS, et al. Vitamin D, calcium, and dairy intakes and stress fractures among female adolescents. *Arch Pediatr Adolesc Med.* 2012;166:595–600.
- Varley I, Hughes DC, Greeves JP, et al. RANK/RANKL/OPG pathway: genetic associations with stress fracture period prevalence in elite athletes. *Bone.* 2015;71:131–136.
- Yanovich R, Milgrom R, Friedman E, et al. Androgen receptor CAG repeat size is associated with stress fracture risk: a pilot study. *Clin Orthop Relat Res.* 2011;469:2925–2931.
- Yanovich R, Friedman E, Milgrom R, et al. Candidate gene analysis in Israeli soldiers with stress fractures. *J Sports Sci Med.* 2012;11:147–155.
- Korvala J, Hartikka H, Pihlajamäki H, et al. Genetic predisposition for femoral neck stress fractures in military conscripts. *BMC Genet.* 2010;11:95.
- Friedman E, Moran DS, Ben-Avraham D, et al. Novel candidate genes putatively involved in stress fracture predisposition detected by whole-exome sequencing. *Genet Res (Camb).* 2014;96:e004.
- Boyle MJ, Hogue GD, Heyworth BE, et al. Femoral neck stress fractures in children younger than 10 years of age. *J Pediatr Orthop.* 2017;37:e96–e99.
- Moo IH, Lee YH, Lim KK, et al. Bilateral femoral neck stress fractures in military recruits with unilateral hip pain. *J R Army Med Corps.* 2016;162:387–390.
- Kupfer KR, Bush DM, Cornell JE, et al. Femoral neck stress fracture in Air Force basic trainees. *Mil Med.* 2014;179:56–61.
- Edwards WB, Gillette JC, Thomas JM, et al. Internal femoral forces and moments during running: implications for stress fracture development. *Clin Biomech (Bristol, Avon).* 2008;23:1269–1278.
- Kuhn KM, Riccio AI, Saldúa NS, et al. Acetabular retroversion in military recruits with femoral neck stress fractures. *Clin Orthop Relat Res.* 2010;468:846–851.
- Goldin M, Anderson CN, Fredericson M, et al. Femoral neck stress fractures and imaging features of femoroacetabular impingement. *PM R.* 2015;7:584–592.

39. Carey T, Key C, Oliver D, et al. Prevalence of radiographic findings consistent with femoroacetabular impingement in military personnel with femoral neck stress fractures. *J Surg Orthop Adv.* 2013;22:54–58.
40. May LA, Chen DC, Bui-Mansfield LT, et al. Rapid magnetic resonance imaging evaluation of femoral neck stress fractures in a U.S. active duty military population. *Mil Med.* 2017;182:e1619–e1625.
41. Bryant LR, Song WS, Banks KP, et al. Comparison of planar scintigraphy alone and with SPECT for the initial evaluation of femoral neck stress fracture. *AJR Am J Roentgenol.* 2008;191:1010–1015.
42. Chalupa RL, Rivera JC, Tennent DJ, et al. Correlation between femoral neck shaft angle and surgical management in trainees with femoral neck stress fractures. *US Army Med Dep J.* 2016;1–5.
43. Feldman JJ, Bowman EN, Phillips BB, et al. Tibial stress fractures in athletes. *Orthop Clin North Am.* 2016;47:733–741.
44. Meardon SA, Willson JD, Gries SR, et al. Bone stress in runners with tibial stress fracture. *Clin Biomech (Bristol, Avon).* 2015;30:895–902.
45. Xu C, Silder A, Zhang J, et al. A cross-sectional study of the effects of load carriage on running characteristics and tibial mechanical stress: implications for stress-fracture injuries in women. *BMC Musculoskelet Disord.* 2017;18:125.
46. Weidauer LA, Binkley T, Vukovich M, et al. Greater polar moment of inertia at the tibia in athletes who develop stress fractures. *Orthop J Sports Med.* 2014;2:2325967114541411.
47. Edwards WB, Taylor D, Rudolph TJ, et al. Effects of running speed on a probabilistic stress fracture model. *Clin Biomech (Bristol, Avon).* 2010;25:372–377.
48. Mallee WH, Weel H, van Dijk CN, et al. Surgical versus conservative treatment for high-risk stress fractures of the lower leg (anterior tibial cortex, navicular and fifth metatarsal base): a systematic review. *Br J Sports Med.* 2015;49:370–376.
49. Cruz AS, de Hollanda JP, Duarte A Jr, et al. Anterior tibial stress fractures treated with anterior tension band plating in high-performance athletes. *Knee Surg Sports Traumatol Arthrosc.* 2013;21:1447–1450.
50. Zbeda RM, Sculco PK, Urch EY, et al. Tension band plating for chronic anterior tibial stress fractures in high-performance athletes. *Am J Sports Med.* 2015;43:1712–1718.
51. Bergstra SA, Kluitenberg B, Dekker R, et al. Running with a minimalist shoe increases plantar pressure in the forefoot region of healthy female runners. *J Sci Med Sport.* 2015;18:463–468.
52. Fujitaka K, Taniguchi A, Isomoto S, et al. Pathogenesis of fifth metatarsal fractures in college soccer players. *Orthop J Sports Med.* 2015;3:2325967115603654.
53. Banal F, Gandjbakhch F, Foltz V, et al. Sensitivity and specificity of ultrasonography in early diagnosis of metatarsal bone stress fractures: a pilot study of 37 patients. *J Rheumatol.* 2009;36:1715–1719.
54. Chuckpaiwong B, Queen RM, Easley ME, et al. Distinguishing Jones and proximal diaphyseal fractures of the fifth metatarsal. *Clin Orthop Relat Res.* 2008;466:1966–1970.
55. Yates J, Feeley I, Sasikumar S, et al. Jones fracture of the fifth metatarsal: Is operative intervention justified? A systematic review of the literature and meta-analysis of results. *Foot (Edinb).* 2015;25:251–257.
56. Porter DA, Rund AM, Dobslaw R, et al. Comparison of 4.5- and 5.5-mm cannulated stainless steel screws for fifth metatarsal Jones fracture fixation. *Foot Ankle Int.* 2009;30:27–33.
57. Ochenjele G, Ho B, Switaj PJ, et al. Radiographic study of the fifth metatarsal for optimal intramedullary screw fixation of Jones fracture. *Foot Ankle Int.* 2015;36:293–301.
58. Lee KT, Park YU, Jegal H, et al. Prognostic classification of fifth metatarsal stress fracture using plantar gap. *Foot Ankle Int.* 2013;34:691–696.
59. Williams DS 3rd, McClay IS, Hamill J. Arch structure and injury patterns in runners. *Clin Biomech (Bristol, Avon).* 2001;16:341–347.
60. Clutton J, Perera A. Vitamin D insufficiency and deficiency in patients with fractures of the fifth metatarsal. *Foot (Edinb).* 2016;27:50–52.
61. Lee KT, Park YU, Young KW, et al. The plantar gap: another prognostic factor for fifth metatarsal stress fracture. *Am J Sports Med.* 2011;39:2206–2211.
62. Wright RW, Fischer DA, Shively RA, et al. Refracture of proximal fifth metatarsal (Jones) fracture after intramedullary screw fixation in athletes. *Am J Sports Med.* 2000;28:732–736.
63. Lee KT, Park YU, Jegal H, et al. Factors associated with recurrent fifth metatarsal stress fracture. *Foot Ankle Int.* 2013;34:1645–1653.
64. Nagel A, Fernholz F, Kibele C, et al. Long distance running increases plantar pressures beneath the metatarsal heads: a barefoot walking investigation of 200 marathon runners. *Gait Posture.* 2008;27:152–155.
65. Hossain M, Clutton J, Ridgwell M, et al. Stress fractures of the foot. *Clin Sports Med.* 2015;34:769–790.
66. Theodorou DJ, Theodorou SJ, Boutin RD, et al. Stress fractures of the lateral metatarsal bones in metatarsus adductus foot deformity: a previously unrecognized association. *Skeletal Radiol.* 1999;28:679–684.
67. Rongstad KM, Tuetting J, Rongstad M, et al. Fourth metatarsal base stress fractures in athletes: a case series. *Foot Ankle Int.* 2013;34:962–968.
68. Gross CE, Nunley JA 2nd. Navicular stress fractures. *Foot Ankle Int.* 2015;36:1117–1122.
69. Shakked RJ, Walters EE, O'Malley MJ. Tarsal navicular stress fractures. *Curr Rev Musculoskelet Med.* 2017;10:122–130.
70. Mann JA, Pedowitz DI. Evaluation and treatment of navicular stress fractures, including nonunions, revision surgery, and persistent pain after treatment. *Foot Ankle Clin.* 2009;14:187–204.
71. McKeon KE, McCormick JJ, Johnson JE, et al. Intraosseous and extraosseous arterial anatomy of the adult navicular. *Foot Ankle Int.* 2012;33:857–861.
72. Torg JS, Moyer J, Gaughan JP, et al. Management of tarsal navicular stress fractures: conservative versus surgical treatment: a meta-analysis. *Am J Sports Med.* 2010;38:1048–1053.
73. Khan KM, Fuller PJ, Brukner PD, et al. Outcome of conservative and surgical management of navicular stress fracture in athletes. Eighty-six cases proven with computerized tomography. *Am J Sports Med.* 1992;20:657–666.
74. Malliaropoulos N, Alaseirli D, Konstantinidis G, et al. Therapeutic ultrasound in navicular stress injuries in elite track and field athletes. *Clin J Sport Med.* 2017;27:278–282.
75. McCormick JJ, Bray CC, Davis WH, et al. Clinical and computed tomography evaluation of surgical outcomes in

- tarsal navicular stress fractures. *Am J Sports Med.* 2011;39:1741–1748.
76. Coulibaly MO, Jones CB, Sietsema DL, et al. Results and complications of operative and non-operative navicular fracture treatment. *Injury.* 2015;46:1669–1677.
 77. Nitta A, Sakai T, Goda Y, et al. Prevalence of symptomatic lumbar spondylolysis in pediatric patients. *Orthopedics.* 2016;39:e434–e437.
 78. Kaneko H, Murakami M, Nishizawa K. Prevalence and clinical features of sports-related lumbosacral stress injuries in the young. *Arch Orthop Trauma Surg.* 2017;137:685–691.
 79. Tezuka F, Sairyo K, Sakai T, et al. Etiology of adult-onset stress fracture in the lumbar spine. *Clin Spine Surg.* 2017;30:E233–E238.
 80. Lawrence JP, Greene HS, Grauer JN. Back pain in athletes. *J Am Acad Orthop Surg.* 2006;14:726–735.
 81. Ledonio CG, Burton DC, Crawford CH 3rd, et al. Current evidence regarding diagnostic imaging methods for pediatric lumbar spondylolysis: a report from the Scoliosis Research Society Evidence-Based Medicine Committee. *Spine Deform.* 2017;5:97–101.
 82. Rush JK, Astur N, Scott S, et al. Use of magnetic resonance imaging in the evaluation of spondylolysis. *J Pediatr Orthop.* 2015;35:271–275.
 83. Steiner ME, Micheli LJ. Treatment of symptomatic spondylolysis and spondylolisthesis with the modified Boston brace. *Spine.* 1985;10:937–943.
 84. Sakai T, Tezuka F, Yamashita K, et al. Conservative treatment for bony healing in pediatric lumbar spondylolysis. *Spine.* 2016.
 85. Scheepers MS, Streak Gomersall J, Munn Z. The effectiveness of surgical versus conservative treatment for symptomatic unilateral spondylolysis of the lumbar spine in athletes: a systematic review. *JBI Database System Rev Implement Rep.* 2015;13:137–173.
 86. Cook RW, Hsu WK. Return to play after lumbar spine surgery. *Clin Sports Med.* 2016;35:609–619.
 87. Miller TL, Kaeding CC. Upper-extremity stress fractures: distribution and causative activities in 70 patients. *Orthopedics.* 2012;35:789–793.
 88. Chaudhury S, Hobart SJ, Rodeo SA. Bilateral first rib stress fractures in a female swimmer: a case report. *J Shoulder Elbow Surg.* 2012;21:e6–e10.
 89. Coris EE, Higgins HW 2nd. First rib stress fractures in throwing athletes. *Am J Sports Med.* 2005;33:1400–1404.
 90. Eng J, Westcott J, Better N. Stress fracture of the first rib in a weightlifter. *Clin Nucl Med.* 2008;33:371–373.
 91. Jones GL. Upper extremity stress fractures. *Clin Sports Med.* 2006;25:159–174, xi.
 92. McDonnell LK, Hume PA, Nolte V. Rib stress fractures among rowers: definition, epidemiology, mechanisms, risk factors and effectiveness of injury prevention strategies. *Sports Med.* 2011;41:883–901.
 93. McKenzie DC. Stress fracture of the rib in an elite oarsman. *Int J Sports Med.* 1989;10:220–222.
 94. Prisk VR, Hamilton WG. Stress fracture of the first rib in weight-trained dancers. *Am J Sports Med.* 2008;36:2444–2447.
 95. Banks KP, Ly JQ, Beall DP, et al. Overuse injuries of the upper extremity in the competitive athlete: magnetic resonance imaging findings associated with repetitive trauma. *Curr Probl Diagn Radiol.* 2005;34:127–142.
 96. Anderson MW. Imaging of upper extremity stress fractures in the athlete. *Clin Sports Med.* 2006;25:489–504, vii.
 97. Aitken SA, Watson BS, Wood AM, et al. Sports-related fractures in South East Scotland: an analysis of 990 fractures. *J Orthop Surg (Hong Kong).* 2014;22:313–317.
 98. Court-Brown CM, Wood AM, Aitken S. The epidemiology of acute sports-related fractures in adults. *Injury.* 2008;39:1365–1372.
 99. Postacchini F, Gumina S, De Santis P, et al. Epidemiology of clavicle fractures. *J Shoulder Elbow Surg.* 2002;11:452–456.
 100. Robinson CM. Fractures of the clavicle in the adult. Epidemiology and classification. *J Bone Joint Surg Br.* 1998;80:476–484.
 101. Neer CS 2nd. Nonunion of the clavicle. *J Am Med Assoc.* 1960;172:1006–1011.
 102. Rowe CR. An atlas of anatomy and treatment of midclavicular fractures. *Clin Orthop Relat Res.* 1968;58:29–42.
 103. Hill JM, McGuire MH, Crosby LA. Closed treatment of displaced middle-third fractures of the clavicle gives poor results. *J Bone Joint Surg Br.* 1997;79:537–539.
 104. McKee MD, Pedersen EM, Jones C, et al. Deficits following nonoperative treatment of displaced midshaft clavicular fractures. *J Bone Joint Surg Am.* 2006;88:35–40.
 105. Morgan RJ, Bankston LS Jr, Hoenig MP, et al. Evolving management of middle-third clavicle fractures in the National Football League. *Am J Sports Med.* 2010;38:2092–2096.
 106. Nowak J, Holgersson M, Larsson S. Sequelae from clavicular fractures are common: a prospective study of 222 patients. *Acta Orthop.* 2005;76:496–502.
 107. Altamimi SA, McKee MD, Canadian Orthopaedic Trauma S. Nonoperative treatment compared with plate fixation of displaced midshaft clavicular fractures. Surgical technique. *J Bone Joint Surg Am.* 2008;90 Suppl 2 Pt 1:1–8.
 108. McKee RC, Whelan DB, Schemitsch EH, et al. Operative versus nonoperative care of displaced midshaft clavicular fractures: a meta-analysis of randomized clinical trials. *J Bone Joint Surg Am.* 2012;94:675–684.
 109. Robertson GA, Wood AM. Return to sport following clavicle fractures: a systematic review. *Br Med Bull.* 2016;119:111–128.
 110. Vander Have KL, Perdue AM, Caird MS, et al. Operative versus nonoperative treatment of midshaft clavicle fractures in adolescents. *J Pediatr Orthop.* 2010;30:307–312.
 111. Sinha AK, Kaeding CC, Wadley GM. Upper extremity stress fractures in athletes: clinical features of 44 cases. *Clin J Sport Med.* 1999;9:199–202.
 112. Ranalletta M, Rossi LA, Piuze NS, et al. Return to sports after plate fixation of displaced midshaft clavicular fractures in athletes. *Am J Sports Med.* 2015;43:565–569.
 113. Meisterling SW, Cain EL, Fleisig GS, et al. Return to athletic activity after plate fixation of displaced midshaft clavicle fractures. *Am J Sports Med.* 2013;41:2632–2636.
 114. Kaplan LD, Flanagan DC, Norwig J, et al. Prevalence and variance of shoulder injuries in elite collegiate football players. *Am J Sports Med.* 2005;33:1142–1146.
 115. Stanley D, Trowbridge EA, Norris SH. The mechanism of clavicular fracture. A clinical and biomechanical analysis. *J Bone Joint Surg Br.* 1988;70:461–464.
 116. Creighton DW, Shrier I, Shultz R, et al. Return-to-play in sport: a decision-based model. *Clin J Sport Med.* 2010;20:379–385.
 117. Cascio BM, Culp L, Cosgarea AJ. Return to play after anterior cruciate ligament reconstruction. *Clin Sports Med.* 2004;23:395–408, ix.
 118. Myer GD, Paterno MV, Ford KR, et al. Neuromuscular training techniques to target deficits before return to sport after

- anterior cruciate ligament reconstruction. *J Strength Cond Res.* 2008;22:987–1014.
119. Orchard J, Best TM, Verrall GM. Return to play following muscle strains. *Clin J Sport Med.* 2005;15:436–441.
120. Matheson GO, Shultz R, Bido J, et al. Return-to-play decisions: are they the team physician's responsibility? *Clin J Sport Med.* 2011;21:25–30.
121. Shrier I. Strategic Assessment of Risk and Risk Tolerance (StARRT) framework for return-to-play decision-making. *Br J Sports Med.* 2015;49:1311–1315.
122. Shultz R, Bido J, Shrier I, et al. Team clinician variability in return-to-play decisions. *Clin J Sport Med.* 2013;23:456–461.